

You MUST lay out your work clearly and concisely and, in this topic more than ever, must not cut corners. Everything must be explained clearly and you must be suitably non-definite in your conclusions.

1. An article in a newspaper claimed that only 40% of people agree with proposals for a new by-pass. A sceptical reader thought that the true figure was more than 40%. She asked 10 friends, 7 of whom agreed with the proposals.
 - (a) Write down suitable null and alternative hypotheses for this test.
 - (b) What is the critical region at a 10% significance level?
 - (c) What is the conclusion of the reader's test?
 - (d) Comment on the validity of this test in the context of this particular sample.
2. The standard drug for treating a certain medical condition causes serious side effects with 40% of patients. A new drug is introduced which, according to its manufacturer, gives a reduced risk of serious side effects.
 - (a) State suitable null and alternative hypotheses to test the manufacturer's claim.
 - (b) A hospital consultant agrees to use the new drug to treat a random sample of 100 patients, all of whom consent. The consultant decides to accept the manufacturer's claim if fewer than 26 patients suffer serious side effects.
 - (i) State the critical region for this test.
 - (ii) Suggest the possible significance level that the consultant is using in this case.
3. Previous attempts to treat a particular disease have had a 25% success rate. A drugs company claim that their new drug will have a better success rate.
 - (a) Write down suitable null and alternative hypotheses for a test.
 - (b) In trials using this new drug, 18 out of 50 patients with the disease recovered. Carry out the hypothesis test at the 1% level of significance. Give your conclusion in words.
4. A firm producing mugs has a quality control scheme in which a random sample of 10 mugs from each batch is inspected. For 50 such samples, the numbers of defective mugs are as follows:

Number of defective mugs	0	1	2	3	4	5	6+
Number of samples	5	13	15	12	4	1	0

- (a) Find the mean number of defective mugs per sample.
- (b) Show that a reasonable estimate for p , the probability that a mug is defective, is 0.2. Use this figure to calculate the probability that a randomly chosen sample will contain exactly two defective mugs. Comment on the agreement between this value and the observed data.

The management is not satisfied with 20% of the mugs being defective, and introduces a new process to reduce the proportion of defective mugs.

- (c) A random sample of 20 mugs, produced by the new process, contains just one which is defective. Test at the 5% level of significance, whether it is reasonable to suppose that the proportion of defective mugs has been reduced, stating your null and alternative hypotheses clearly.
- (d) What would the conclusion have been if the management had chosen to conduct the test at the 10% level?

5. A newspaper article claims that in newly married couples the husband is equally likely to be younger or older than the wife. In the past it was far more likely that the husband was older than the wife, so a vicar thinks that this may not be true.

(e) Letting p denote the probability that the husband is older than the wife, write down the null and alternative hypotheses which the vicar should use.

In the next month the vicar marries 29 couples, and finds that in 23 of these marriages the husband is older than the wife.

(f) Assuming the null hypothesis is true, find the probability of

- (i) exactly 23 couples with the husband older than the wife;
- (ii) 23 or more couples with the husband older than the wife.

(g) The vicar carries out his test at the 5% level. What conclusion should he reach?